

PSEUDOCODE WRITING

← Back

QUESTIONS

Q1 ✓
Student Management System
[BL3 – Apply]
10 marks

Q2 ✓
Library Book Management [BL3 – Apply]
10 marks

Q3 ✓
Employee Salary Calculation
[BL3 – Apply]
10 marks

Q4 ✓
Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks

Q5 ✓
5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q6 ✓

Q7 7. Hospital Patient Record System [BL4 – Analyze]

10 marks

Design a Patient base class and analyze how it extends into:

- InPatient — with ward number, admission date, and daily charges
- OutPatient — with appointment date and consultation fee

Analyze encapsulation requirements for sensitive data (diagnosis, prescription). Override a generateBill() method for each type and compare the billing logic differences. OOP Concepts: Encapsulation | Inheritance | Polymorphism

Your Answer

```
import java.time.LocalDate;

class Patient {
    private String patientId;
    private String patientName;
    private String diagnosis;
    private String prescription;

    public Patient(String patientId, String patientName, String diagnosis, String prescription) {
        this.patientId = patientId;
        this.patientName = patientName;
    }
}
```

Q3

Employee Salary Calculation
[BL3 – Apply]
10 marks

Q4

Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks

Q5

5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q6

6. Bank Account Hierarchy [BL4
– Analyze]
10 marks

Q7

7. Hospital Patient Record
System [BL4 – Analyze]
10 marks

Q8

8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

Q9

Your Answer

```
this.patientName = patientName;
this.diagnosis = diagnosis;
this.prescription = prescription;
}

public String getPatientId() {
    return patientId;
}

public String getPatientName() {
    return patientName;
}

public String getDiagnosis() {
    return diagnosis;
}

public void setDiagnosis(String diagnosis) {
    this.diagnosis = diagnosis;
}

public String getPrescription() {
    return prescription;
}
```

Q3

Employee Salary Calculation

[BL3 – Apply]

10 marks



Q4

Shape Area Calculator using

Polymorphism [BL3 – Apply]

10 marks



Q5

5. Vehicle Registration System

[BL3 – Apply]

10 marks



Q6

6. Bank Account Hierarchy [BL4

– Analyze]

10 marks



Q7

7. Hospital Patient Record

System [BL4 – Analyze]

10 marks



Q8

8. E-Commerce Product Catalog

[BL4 – Analyze]

10 marks



Q9



Your Answer

```
public String getPrescription() {  
    return prescription;  
}  
  
public void setPrescription(String prescription) {  
    this.prescription = prescription;  
}  
  
public double generateBill() {  
    return 0.0;  
}  
}  
  
class InPatient extends Patient {  
    private int wardNumber;  
    private LocalDate admissionDate;  
    private double dailyCharges;  
    private int numberOfDays;  
  
    public InPatient(String patientId, String patientName, String diagnosis, String prescription,  
        int wardNumber, LocalDate admissionDate, double dailyCharges, int numberOfDays) {  
        super(patientId, patientName, diagnosis, prescription);  
        this.wardNumber = wardNumber;  
        this.admissionDate = admissionDate;  
        this.dailyCharges = dailyCharges;  
    }  
}
```


Q3 ✓
Employee Salary Calculation
[BL3 – Apply]
10 marks

Q4 ✓
Shape Area Calculator using
Polymorphism [BL3 – Apply]
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5. Vehicle Registration System
[BL3 – Apply]
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– Analyze]
10 marks

Q7 ✓
7. Hospital Patient Record
System [BL4 – Analyze]
10 marks

Q8 ✓
8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

Q9 ✓

Your Answer

```
this.dailyCharges = dailyCharges;
this.numberOfDays = numberOfDays;
}

public int getWardNumber() {
    return wardNumber;
}

public LocalDate getAdmissionDate() {
    return admissionDate;
}

public double getDailyCharges() {
    return dailyCharges;
}

public int getNumberOfDays() {
    return numberOfDays;
}

@Override
public double generateBill() {
    return dailyCharges * numberOfDays;
}
}
```

Q3

Employee Salary Calculation
[BL3 – Apply]
10 marks



Q4

Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks



Q5

5. Vehicle Registration System
[BL3 – Apply]
10 marks



Q6

6. Bank Account Hierarchy [BL4
– Analyze]
10 marks



Q7

7. Hospital Patient Record
System [BL4 – Analyze]
10 marks



Q8

8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks



Q9



Your Answer

```
        return dailyCharges * numberOfDays;
    }
}

class OutPatient extends Patient {
    private LocalDate appointmentDate;
    private double consultationFee;

    public OutPatient(String patientId, String patientName, String diagnosis, String prescription,
        LocalDate appointmentDate, double consultationFee) {
        super(patientId, patientName, diagnosis, prescription);
        this.appointmentDate = appointmentDate;
        this.consultationFee = consultationFee;
    }

    public LocalDate getAppointmentDate() {
        return appointmentDate;
    }

    public double getConsultationFee() {
        return consultationFee;
    }

    @Override
    public double generateBill() {
```

Employee Salary Calculation
[BL3 – Apply]
10 marks

Q4

Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks

Q5

5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q6

6. Bank Account Hierarchy [BL4
– Analyze]
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Q7

7. Hospital Patient Record
System [BL4 – Analyze]
10 marks

Q8

8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

Q9

9. University Staff Hierarchy
[BL4 – Analyze]

```
public double generateBill() {  
    return consultationFee;  
}  
  
public class HospitalPatientRecordSystem {  
    public static void main(String[] args) {  
        Patient p1 = new InPatient(  
            "P001", "Ravi",  
            "Flu", "Rest + Vitamin C",  
            12, LocalDate.of(2026, 7, 1),  
            500.0, 4  
        );  
  
        Patient p2 = new OutPatient(  
            "P002", "Meena",  
            "Allergy", "Antihistamine",  
            LocalDate.of(2026, 7, 10),  
            300.0  
        );  
  
        System.out.println("InPatient Bill: " + p1.generateBill());  
        System.out.println("OutPatient Bill: " + p2.generateBill());  
    }  
}
```